

CO3 - AT3 Tree Traversal Challenge

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Course Code: CSA0311

Course Name: Data Structures

In order: 10, 20, 30, 100, 150, 200, 300
 Left Root Right

Post order: 100, 20, 10, 30, 200, 150, 300
 Root Left Right

Pre order: 10, 30, 20, 150, 300, 200, 100
 Left Right Root

In order: 10, 15, 20, 23, 25, 30, 35, 39, 42
 Left Root Right

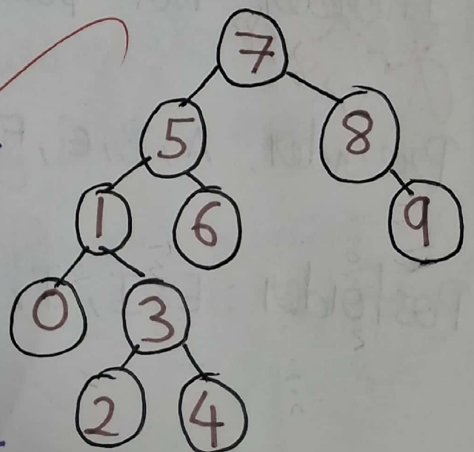
Pre order: 30, 20, 10, 15, 25, 23, 39, 35, 42
 Root Left Right

Post order: 15, 10, 23, 25, 20, 35, 42, 39, 30
 Left Right Root

In order: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
 Left Root Right

Pre order: 7, 5, 1, 0, 3, 2, 4, 6, 8, 9
 Root Left Right

Post order: 0, 2, 4, 3, 1, 6, 5, 9, 8, 7
 Left Right Root



6) Inorder: 2, 3, 4, 6, 7, 9, 13, 15, 17, 1, 20
 Left Root Right
 Preorder: 15, 6, 3, 2, 4, 7, 13, 9, 18, 7, 20
 Root Left Right
 Postorder: 2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15
 Left Right Root

5) Inorder: 7, 4, 2, 5, 1, 3, 6
 Left Root Right
 Preorder: 1, 2, 4, 7, 5, 3, 6
 Root Left Right
 Postorder: 7, 4, 5, 2, 6, 3, 1
 Left Right Root

7) Inorder traversal is (3) A, B, C, D, E, F, G, H, I

8) Postorder Traversal is (3) D, F, E, B, H, G, C, A

9) Inorder: 4, 6, 3, 7, 2, 5, 1
 Left Root Right
 Preorder: 7, 6, 4, 3, 5, 2, 1
 Root Left Right
 Postorder: 4, 3, 6, 2, 1, 5, 7
 Left Right Root

10) Inorder: Not possible, since A has two children

Preorder: A, B, E, F, I, J, K, C, G, H, D (Root → children)

Postorder: E, I, J, K, F, B, G, H, C, D, A (Children → Root)